

Haliç University Management Information Systems

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ADEA CAR RENTAL SYSTEM – MILESTONE 1 REPORT

Section 1: Objective and Scope The primary objective of this milestone was to establish the foundational architecture for the ADEA Car Rental System. The scope included implementing a role-based routing mechanism, configuring global state management for user sessions, and successfully integrating a live cloud-based NoSQL database to retrieve and display vehicle inventory dynamically.

Section 2: Individual Contribution & Implementation Several key technical components were developed and integrated:

- **Architectural Organization:** All website files were organized under a single root folder with semantic naming conventions applied across all directories.
- **Routing and Authentication Logic:** A functional navigation module with three distinct user roles (Admin, Rental Agent, Customer) was implemented using the React Router library. A programmatic redirection algorithm routes users to their respective dashboards.
- **State Management:** A global state container was established utilizing the Zustand library to maintain session persistence and manage authentication status.
- **Database Integration:** A live cloud database was successfully integrated using Google Firebase Firestore. Asynchronous functions were implemented within the Customer dashboard to fetch and display real-time vehicle data from the "cars" collection.

Section 3: Visual Evidence of Implementation

The following images demonstrate the successful integration of the live database and the role-based dashboard redirection:

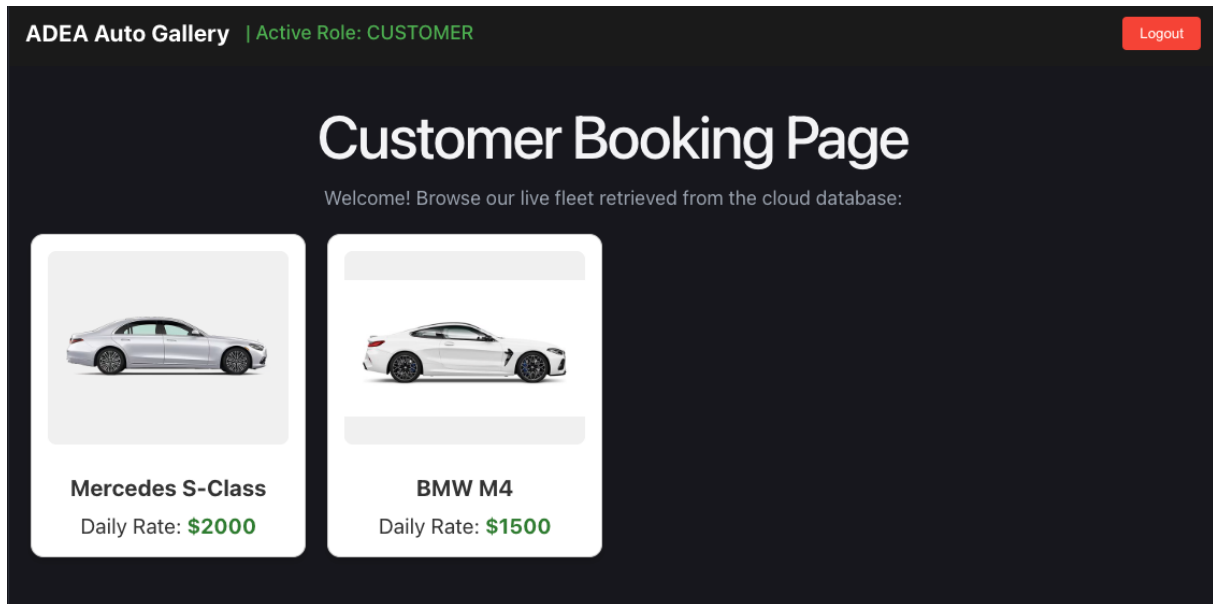
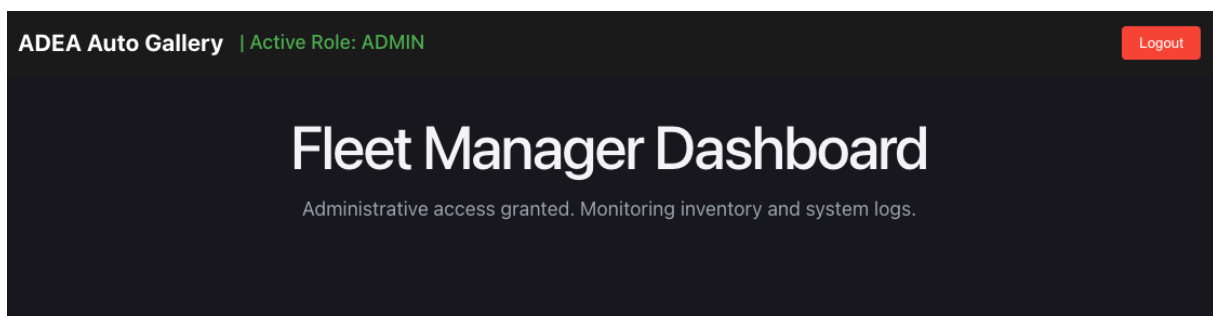
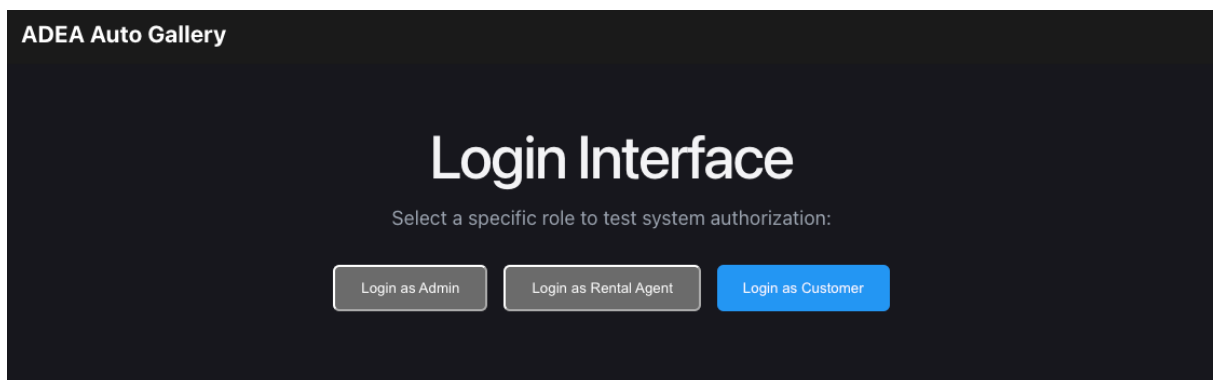


Figure 1: Customer Booking Page displaying real-time vehicle data (Brand, Model, Price, and Image) retrieved from Firebase Firestore.



Rental Agent Interface

Operational access granted. Managing check-in/check-out workflow.

Figure 2: Successful programmatic redirection to the Role-Based Dashboard after authentication.

Section 4: Technical Challenges and Solutions During the database integration phase, a significant configuration challenge was encountered:

- **Problem:** Upon initial setup, the Cloud Firestore "Production Mode" was inadvertently selected, defaulting security rules to allow read, write: if false;, which locked all external access and caused permission errors in the React application.
- **Resolution:** The Firestore "Rules" console was accessed, and the security configuration was manually updated to allow read, write: if true; for the development phase. This resolved the permission conflict and enabled live data retrieval.

Section 5: Conclusion The technical requirements for Milestone 1 were successfully met. The application now features a secure routing mechanism and an active connection to a live database, providing a solid foundation for the subsequent development phases of the car rental system.